

Week 03 Task 8

Cyber Kill Chain + MITRE ATT&CK Intro

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1) Executive Summary

This report presents the work completed for Week 03 Task 8, which introduces two important cybersecurity frameworks used to understand and analyse cyber attacks: the Lockheed Martin Cyber Kill Chain and MITRE ATT&CK. The report first explains the seven phases of the Cyber Kill Chain: Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command and Control, and Actions on Objective. It shows how an attack progresses step by step and how defenders can stop it by breaking any single link. It then covers the MITRE ATT&CK framework, including its tactics, techniques and procedures (TTPs), and reviews the Enterprise ATT&CK tactics column to understand the different objectives an attacker may pursue.

To apply this knowledge, the report studies the technique **T1110 Brute Force**, covering its common variations and the indicators a SOC analyst can use to detect it. The same brute-force scenario is then mapped onto each phase of the Cyber Kill Chain, along with detection and defense measures for every phase. Since the TryHackMe MITRE room was premium, the ATT&CK Fundamentals training on the official MITRE website was completed instead.

Overall, this task builds a strong foundation for analysing real-world attacks, classifying alerts and improving an organization's detection and response capabilities

2) Cyber Kill Chain

Cyber Kill Chain is a cyber security framework developed by Lockheed Martin. A concept inspired by military strategies to identify and mitigate security incidents by breaking down an attack into various stages. An adversary must successfully complete every phase to reach their final goal, meaning security teams can stop an attack by breaking any single link.

2.1) The Seven Phases of the Cyber Kill Chain

Cyber kill chain gives the overview of cyber attacks so that organizations have an understanding of each stage and recover their businesses from attack. Each phase gives the overview of a specific type of attack in the cyber kill chain model. The cyber kill chain is the step-by-step techniques that identify, detects, and stops the vulnerable activity.

2.1.1) Reconnaissance

Reconnaissance is the information-gathering phase where an attacker collects details about the target, such as IP addresses, domains, employees, technologies, network services, and potential vulnerabilities. Information can also be gathered through **OSINT (Open-Source Intelligence)** using publicly available sources such as websites, search engines, social media, and public records.

Reconnaissance is generally divided into two types: **Passive Reconnaissance**, where information is collected from publicly available sources without directly interacting with the target, and **Active Reconnaissance**, where the attacker directly interacts with the target, for example through network scanning, port scanning, or service enumeration.

2.1.2) Weaponization

In the weaponization phase, the attacker prepares a malicious payload or attack method based on the information gathered during reconnaissance. This may involve combining malware, malicious documents, scripts, or exploits with a known vulnerability.

2.1.3) Delivery

Delivery is the phase where the attacker sends or transfers the malicious payload to the target. Common delivery methods include phishing emails, malicious attachments, compromised websites, malicious links, and removable media

2.1.4) Exploitation

During exploitation, the attacker takes advantage of a vulnerability or user action to execute malicious code or gain unauthorized access. Successful exploitation allows the attacker to move further into the targeted system.

2.1.5) Installation

In the installation phase, the attacker installs malware, a backdoor, or another malicious component on the compromised system. This can provide persistent access and allow the attacker to maintain control even after the initial compromise.

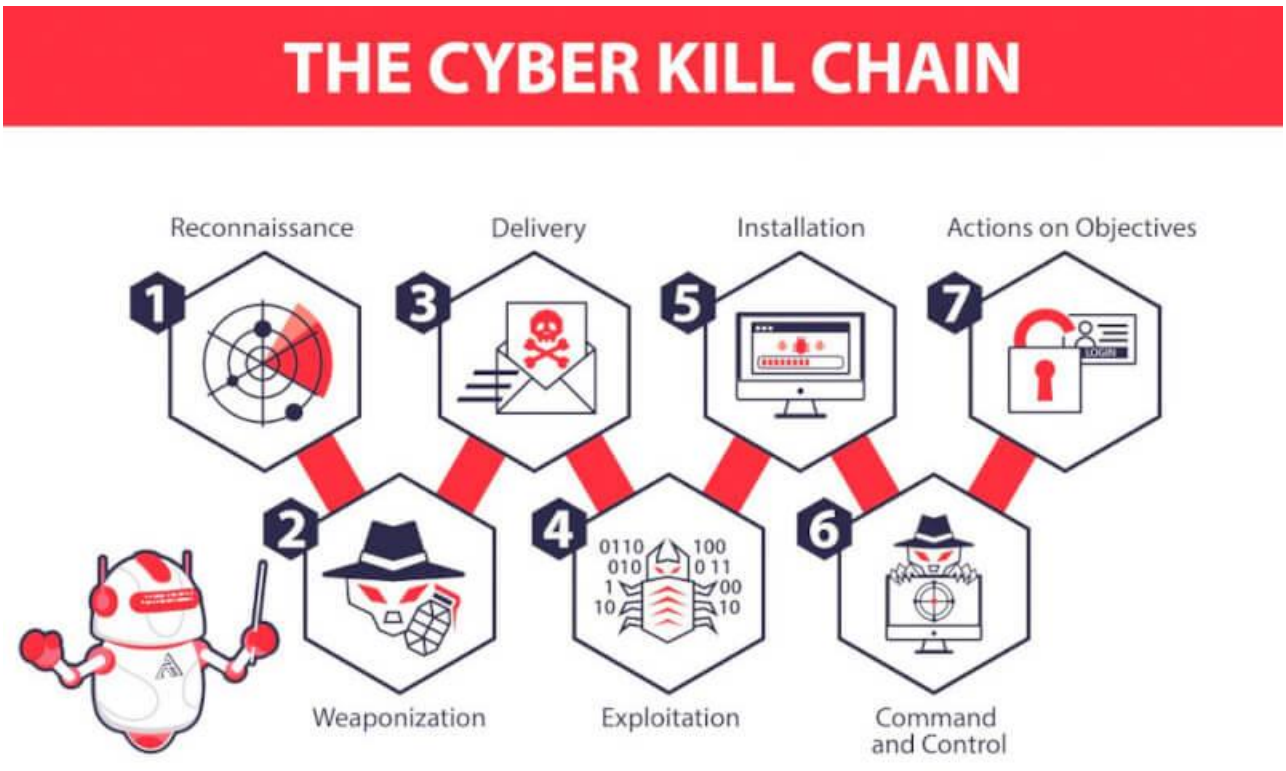
2.1.6) Command and Control (C2)

Command and Control occurs when the compromised system communicates with infrastructure controlled by the attacker. Through this communication channel, the attacker can issue commands, receive information, download additional payloads, and remotely control the infected system

2.1.7) Actions on the Objective

This is the final phase where the attacker performs the main goal of the attack. Depending on the objective, this may include stealing sensitive data, escalating privileges, moving laterally, disrupting services, encrypting files, or exfiltrating information.

PICTORIAL REPRESENTATION OF CYBER KILL CHAIN



3) MITRE Attack

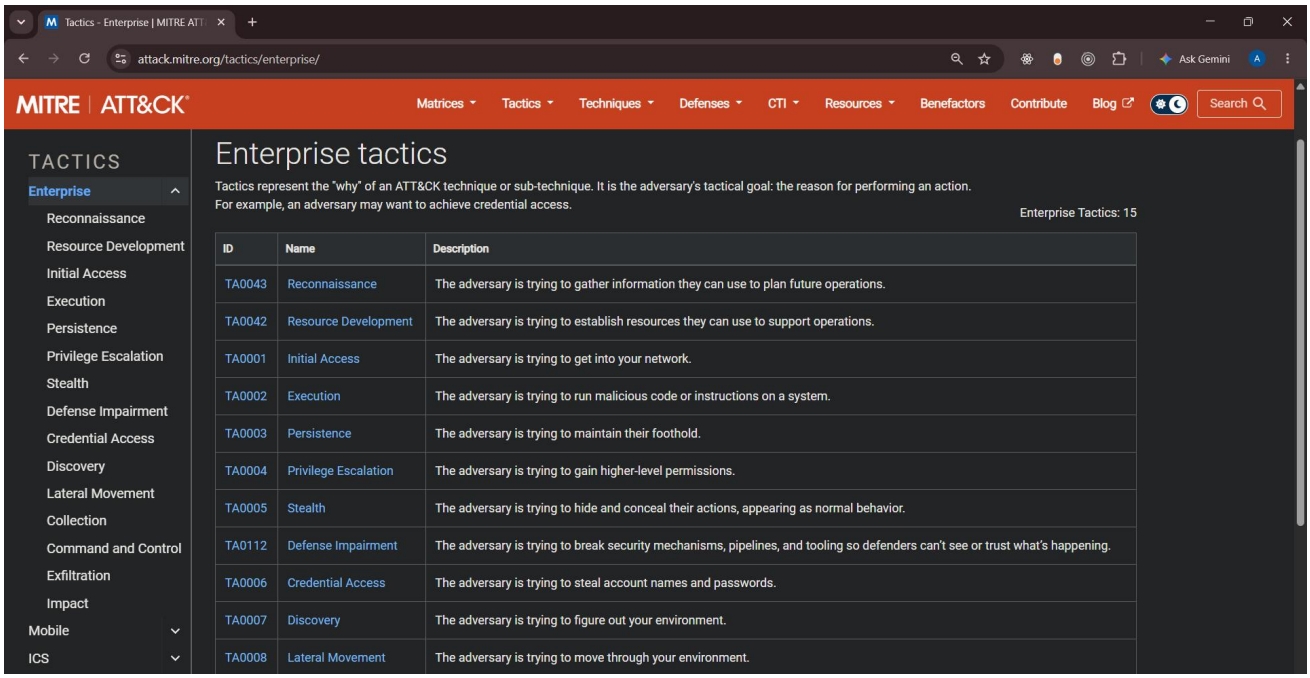
A cyber security framework that is free publicly accessible knowledge base all over the globe maintaining tactics, techniques and Procedures (TTP). It helps security professionals understand how attackers operate throughout different stages of a cyberattack.

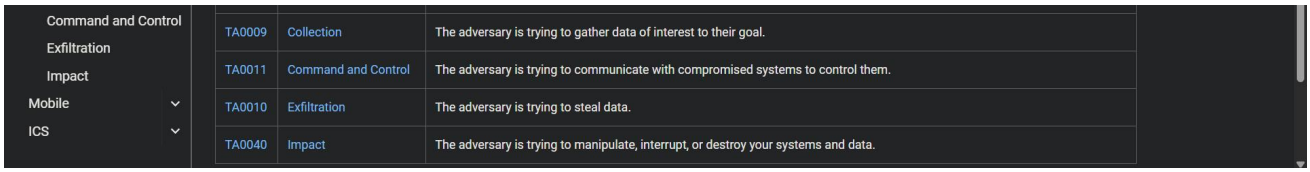
Tactics represent the attacker's objective or **why** an action is performed, such as Initial Access, Persistence, or Credential Access. **Techniques** describe **how** an attacker achieves that objective, while **Procedures** represent the specific methods or tools used by threat actors to implement a technique.

From a **SOC analyst perspective**, MITRE ATT&CK is useful for mapping suspicious activities and security alerts to known attacker techniques. It helps analysts investigate incidents, understand attacker behavior, improve detection rules, perform threat hunting, and identify gaps in an organization's security monitoring.

For example, repeated password attempts against multiple accounts may be mapped to the **Brute Force** technique under the **Credential Access** tactic. This provides analysts with a standardized way to classify and communicate observed malicious behavior.

4) Viewing the Tactics Column





I reviewed the Enterprise ATT&CK Tactics column to understand the different objectives an adversary may attempt to achieve during an attack. The Enterprise ATT&CK matrix currently contains 15 tactics, including Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact.

From a SOC analyst perspective, understanding these tactics helps classify suspicious activities according to the attacker's objective and provides a structured way to understand the progression of an attack.

5) Understanding T1110 Brute Force

T1110 – Brute Force is a MITRE ATT&CK technique used by adversaries to gain unauthorized access by repeatedly attempting to authenticate to an account or system using different passwords, credentials, or authentication methods.

The attacker may use automated tools to perform a large number of login attempts until valid credentials are discovered. Brute-force attacks can target services such as **SSH, RDP, VPNs, web applications, email accounts, and other authentication systems**.

5.1) Common Brute-Force Techniques

- **Password Guessing:** Trying commonly used or likely passwords against an account.
- **Password Cracking:** Attempting to recover passwords from captured password hashes.
- **Password Spraying:** Trying one commonly used password against many different accounts to avoid account lockouts.
- **Credential Stuffing:** Using previously leaked username/password combinations against another service.

5.2) Indicators of Brute force Attack

A SOC analyst may look for:

- Multiple failed login attempts from the same source IP.
- Repeated authentication failures against one account.
- Login attempts against many accounts from one IP.
- Successful authentication immediately following numerous failures.
- Authentication attempts occurring at unusual times or from unusual locations.
- Large numbers of authentication events within a short period.

6) Mapping one Attack onto the kill-chain stages

6.1) Attack Scenario

An attacker discovers a company server with SSH exposed to the internet. Using an automated tool, the attacker tries many passwords until one account with a weak password is compromised. After logging in, the attacker creates persistent access, connects the server to their own infrastructure and copies sensitive data out of the network. Because the attacker must complete every phase to reach the goal, defenders can stop the attack by breaking any single link in this chain.

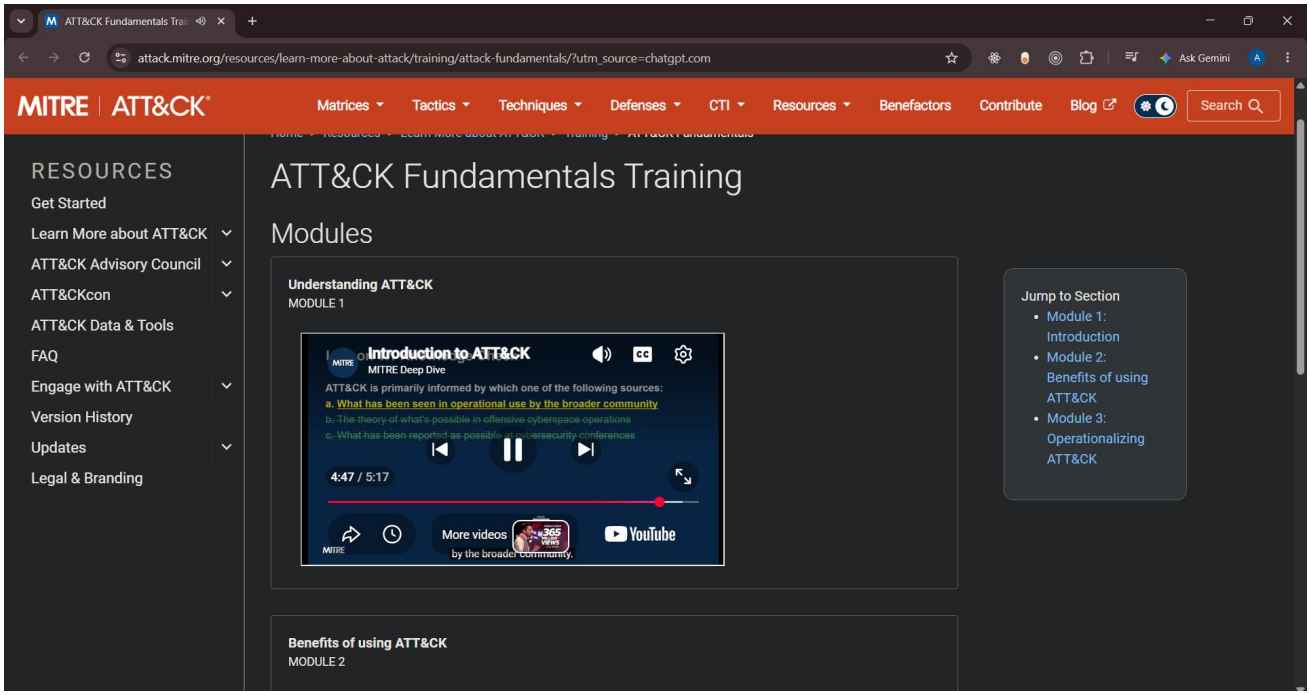
6.2) Mapping an Attack to Kill chain

Kill Chain Phase	Attacke type Brute-force	Detection /Defense
Reconnaissance	The attacker scans the target's public IP range, finds SSH (port 22) open and collects likely usernames from OSINT sources.	Detect port scans and repeated connection attempts from one external IP; block the source.
Weaponization	The attacker prepares a password wordlist	Hard to observe directly; use

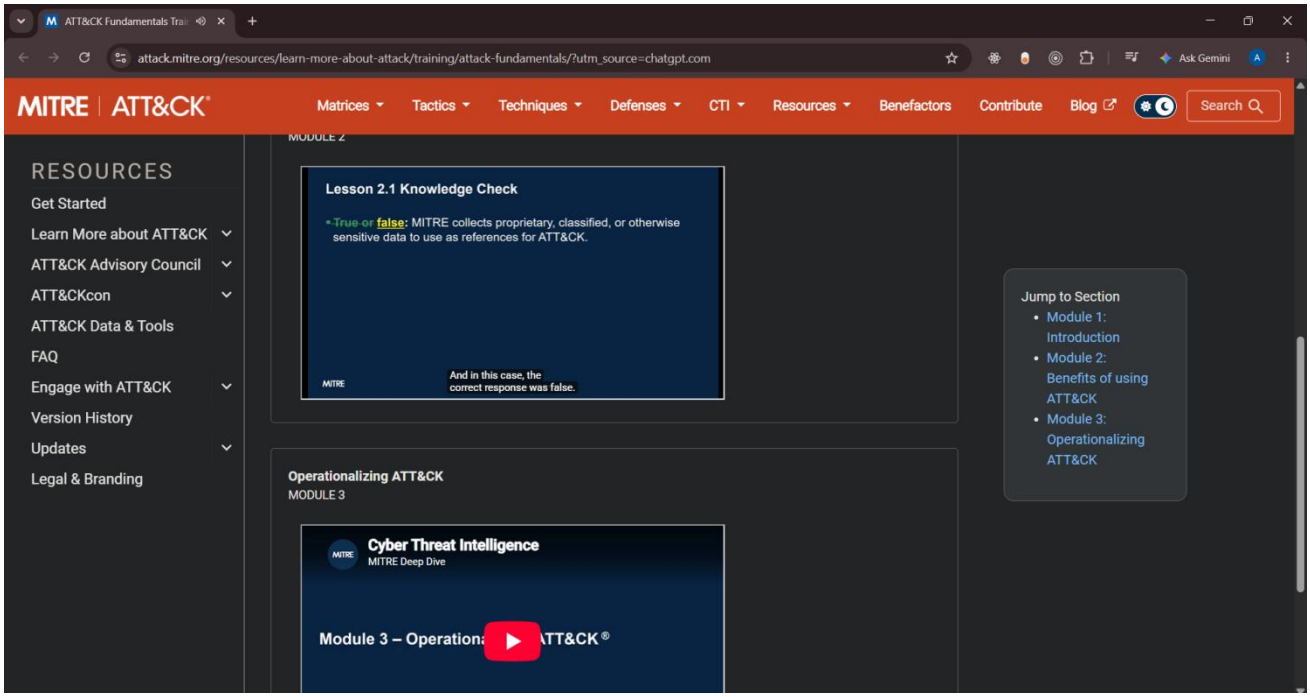
	and configures an automated brute-force tool such as Hydra.	threat intelligence and malicious IP reputation lists.
Delivery	The tool sends a large number of automated login requests to the exposed SSH service.	Monitor authentication logs for bursts of failed logins; restrict SSH with firewall rules or a VPN.
Exploitation	A weak password is guessed and the attacker logs in successfully with a valid account.	Alert on many failures followed by one success; enforce account lockout, strong passwords and MFA.
Installation	The attacker adds an SSH key or creates a new local user to keep access to the server.	Alert on changes to authorized_keys and new account creation; isolate the affected host.
Command and Control	The compromised server downloads tools and opens a remote shell to attacker controlled infrastructure	Inspect outbound traffic for unusual destinations; block malicious domains and IPs.
Actions on Objective	The attacker copies sensitive files from the server and sends them out of the network.	Detect abnormal outbound data volume; reset credentials, restore from backup and investigate the scope.

7) TryHackMe Room MITRE

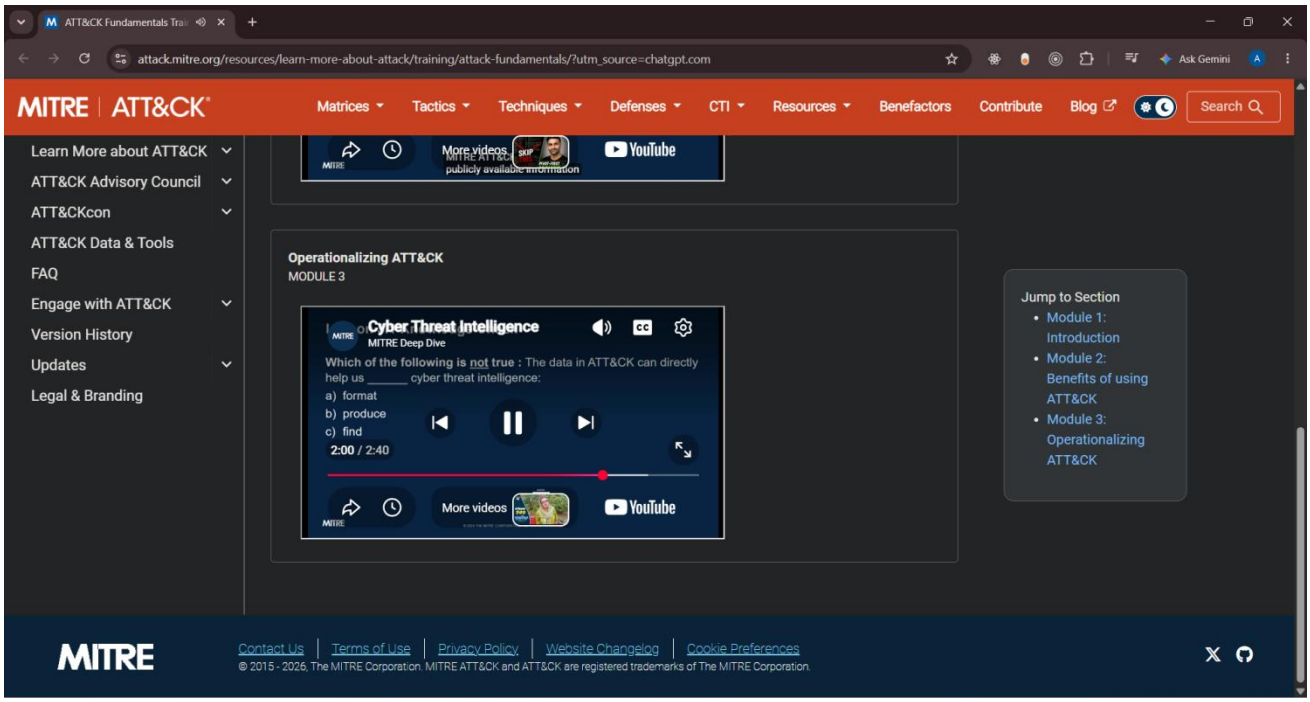
TryHackMe Room MITRE is Premium so that's why I followed the learning resource from MITRE website to learn about MITRE Attack.



In the above figure the 1st module Introduction to Attack is covered. This module introduces what MITRE ATT&CK is and explains that it is built from real-world adversary behavior observed and reported by the wider security community. It gave me the basic foundation of tactics, techniques and how the framework is used to describe attacker behavior.



In the above figure the 2nd module Benefits of using Attack is covered. This module explains why organizations use ATT&CK, such as having a common language for describing attacker behavior and basing defenses on publicly available, real-world information. The knowledge check also showed that MITRE relies only on public data and does not collect proprietary or classified information.



In the above figure the 3rd module of MITRE ATTACK Operationalizing Attack is covered. This module shows how ATT&CK is put into practice, especially in Cyber Threat Intelligence, where its data helps to find, produce and format threat intelligence. It helped me understand how analysts map observed activity to known techniques to improve detection and reporting.

8) Conclusion

In this task, I learned about two important cybersecurity frameworks: the Cyber Kill Chain and **MITRE ATT&CK**. The **Cyber Kill Chain** breaks an attack into seven phases, which helps defenders understand how an intrusion progresses and where it can be stopped. MITRE ATT&CK complements this by describing attacker behavior in detail through tactics, techniques and procedures (**TTPs**).

I also learned how SOC analysts use these frameworks for **threat detection** and **monitoring**. By mapping suspicious activity, such as repeated failed logins under **T1110 Brute Force**, to known attacker techniques, analysts can classify alerts, investigate incidents and communicate findings in a standard way. Studying how attackers operate first makes it possible to build better detection rules, find gaps in monitoring and respond earlier.

Since the TryHackMe MITRE room was premium, I completed the ATT&CK Fundamentals training on the official MITRE website. Overall, this task gave me a solid foundation for analysing real-world attacks and strengthening an organization’s defense.